

Practical-10

Aim : Write a program to implement Bubble Sort.

Input :

```
#include <stdio.h>
```

```
void bubbleSort(int arr[], int n) {  
    int i, j, temp;  
  
    // Bubble Sort  
    for (i = 0; i < n - 1; i++) {  
        for (j = 0; j < n - i - 1; j++) {  
            if (arr[j] > arr[j + 1]) {  
                // Swap elements  
                temp = arr[j];  
                arr[j] = arr[j + 1];  
                arr[j + 1] = temp;  
            }  
        }  
  
        // Print array after each pass  
        printf("After Pass %d: ", i + 1);  
        for (j = 0; j < n; j++) {  
            printf("%d ", arr[j]);  
        }  
        printf("\n");  
    }  
}
```

```
int main() {  
    int arr[100], n, i;  
  
    printf("Enter number of elements: ");  
    scanf("%d", &n);  
  
    printf("Enter elements:\n");  
    for (i = 0; i < n; i++) {  
        scanf("%d", &arr[i]);  
    }  
  
    bubbleSort(arr, n);  
  
    printf("Sorted array: ");  
    for (i = 0; i < n; i++) {  
        printf("%d ", arr[i]);  
    }  
  
    return 0;  
}
```

Output:

```
Enter number of elements: 5  
Enter elements:  
50 70 20 5 100  
After Pass 1: 50 20 5 70 100  
After Pass 2: 20 5 50 70 100  
After Pass 3: 5 20 50 70 100  
After Pass 4: 5 20 50 70 100  
Sorted array: 5 20 50 70 100
```